

# Weekly Progress Reports

FORGE — Functional-posterior Observatory for Refinement and Geometric Exploration

INFO 698 Capstone · University of Arizona · Nathan Herling · Faculty mentor: Prof. Kenneth Johns

Weeks 0–12 · May 18 – Aug 16, 2026 · compiled August 7, 2026

*Each week is reported in three parts — what was accomplished, what caused problems, and what carries into the following week — held to 100 words or fewer. Sources are the project's own `weekly_todos.json` and `gantt.json` records. W12 is stated as planned work, not as an accomplished record.*

## Brainstorming & scope lock

### W0

May 18–24

Intro /  
Planning

#### ACCOMPLISHED THIS WEEK

Project scope brainstormed and locked; TRON Ares documentation-site design started.

#### CAUSED PROBLEMS

Scoping was the whole difficulty — FORGE had to be narrow enough to actually build and broad enough to carry a capstone.

#### NEXT WEEK

Research and planning; stand up the documentation-site scaffold.

## Planning & research

### W1

May 25–31

Intro /  
Planning

#### ACCOMPLISHED THIS WEEK

Research and planning underway; doc-site scaffold and panel router stood up.

#### CAUSED PROBLEMS

None blocking.

#### NEXT WEEK

Video lectures, project commercial, Proposal mk\_1.

## Proposal mk\_1

### W2

Jun 1–7

Intro /  
Planning

#### ACCOMPLISHED THIS WEEK

Doc-site content built out; video lectures recorded; project commercial produced; Proposal mk\_1 submitted.

#### CAUSED PROBLEMS

None blocking.

#### NEXT WEEK

Full proposal document; FORGE splash; lock the timeline and Gantt.

## Planning lock-in

### W3

Jun 8–14

Intro /  
Planning

#### ACCOMPLISHED THIS WEEK

Full proposal submitted; FORGE splash designed; experiments repo scaffolded (per-project skeleton plus shared infra); genre chosen as the first standalone site.

#### CAUSED PROBLEMS

The original timeline did not survive contact and was restructured to an MVP-phase cadence: P1 standalone sites W4–6, P2 integrate and deploy W7–9, stretch W10, risk window W8–10.

#### NEXT WEEK

Music Genre EDA and standalone site; first Bayesian options.

## Music Genre (BEARDOWN)

### W4

Jun 15–21

MVP Phase 1

#### ACCOMPLISHED THIS WEEK

BEARDOWN model 1 live as a local full-stack site; leakage-safe pipeline; backbone frozen at val 0.7467 / test 0.700; seven-file bundle with phi\_train and GGN eigendecomposition pre-staged; compute-cost ledger.

#### CAUSED PROBLEMS

A silent dirty-grey mel failure was scoring dropped songs as noise — fixed by sharing one extraction recipe. The gap to 0.766 was diagnosed to batch size plus run variance; decision was to freeze rather than chase it.

#### NEXT WEEK

Phonon EDA and standalone site.

## Phonon DOS

### W5

Jun 22–28

MVP Phase 1

#### ACCOMPLISHED THIS WEEK

FORGE Bayesian core landed — LLLA plus HMC on the frozen head, with the observatory tab; phonon observatory built end to end on the 51-bin DOS regression.

#### CAUSED PROBLEMS

Entered the week behind schedule. An LLLA intercept bug held phonon at  $R^2 \approx -27$  until DOS baseline centering lifted it to  $\approx 0.95$ . Phonon absorbed the FORGE-layer generalisation, so the work ran into W6.

#### NEXT WEEK

ATLAS EDA and standalone site.

## ATLAS LLP

### W6

Jun 29 – Jul 5

MVP Phase 1

#### ACCOMPLISHED THIS WEEK

Phonon observatory finished with the data-limited verdict, last-layer panel and optimisation frontier; HMC made full-stack with a real server-side sampler; ATLAS skeleton folded in.

#### CAUSED PROBLEMS

ATLAS ran in parallel on the RA track rather than after phonon, and its standalone-site integration slipped. The W6 milestone as worded — three standalone sites — was not met.

#### NEXT WEEK

Integrate the repos; move off localhost.

## Integrate & deploy

### W7

Jul 6–12

MVP Phase 2

#### ACCOMPLISHED THIS WEEK

ATLAS LLP model framing narrowed; external-hosting options surveyed, paid PaaS against tunnel-to-own-hardware.

#### CAUSED PROBLEMS

Light week by circumstance — deliberation, not construction. No code shipped and the entire deployment effort pushed wholesale into W8.

#### NEXT WEEK

Build and deploy the integrated stack.

## Public proof of concept

### W8

Jul 13–19

MVP Phase 2

#### ACCOMPLISHED THIS WEEK

Full stack built — splash, floor, select, dashboard with feature stats, the tau-knob observatory and the Drop Deck — and deployed public through a Cloudflare tunnel off my own hardware, then verified from a remote PC.

#### CAUSED PROBLEMS

The first remote test failed: the serving image never had torch. Fixed with an explicit CPU-torch layer plus an ML-stack smoke gate. Port collisions forced container hosts out of the 5000 range.

#### NEXT WEEK

Begin the write-up; final software edits and testing.

## Front-end restart

### W9

Jul 20–26

MVP Phase 2

#### ACCOMPLISHED THIS WEEK

FORGE contract published; gauges calibrated and locked against real bundles; four tenets mapped to four gauges; model-select deck built; splash rebuilt across three passes; hype video; forge-lite tutorial.

#### CAUSED PROBLEMS

The front end became the binding constraint, so the write-up start was deferred to W10. Drop Deck inference lag was triaged — 14.4 s baseline, the mk3 MC loop 40% of it — but the fixes were parked until the site is up.

#### NEXT WEEK

Write-up; lock the MVP and freeze repos.

## Tutorial re-lock · phonon mk4 opened

### W10

Jul 27 – Aug 2

MVP Phase 3

#### ACCOMPLISHED THIS WEEK

Tutorial locked as the dice phantom after a bench found the first configuration that moves all four gauges at once; dice\_model.py shipped; per-model colours and emblems locked; W9 filed into the docs; research CV built.

#### CAUSED PROBLEMS

Palmer Penguins was rejected, forcing a tutorial restart. Three gauge traps surfaced, each of which silently produces a flat dial. The write-up slipped a second time, W10 to W11.

#### NEXT WEEK

Phonon mk4 from scratch; apply the gauges to the tutorial.

## Metric lock-in · experiments closed

### W11

Aug 3–9

MVP Phase 3

#### ACCOMPLISHED THIS WEEK

Variance locked as the base measure; C1–C4 re-cut; forge-brandeis board built; phonon mk4 closed on a measured accuracy-versus-epistemic-signal frontier; ATLAS three rungs trained; all eight models exported and packaged at 4.54 MB; final report outlined.

#### CAUSED PROBLEMS

The variance switch invalidated the W9 gauge calibration outright. Three phonon trunk retrains were built and refuted. OneDrive silently reverted source files four times, and PowerShell packaging failed twice before moving to Python.

#### NEXT WEEK

Stitch every FORGE component into one site; write the final paper.

## Stitch · submit · present

### W12

Aug 10–16

Deliverables

#### ACCOMPLISHED THIS WEEK

Week not yet run.

#### CAUSED PROBLEMS

Two things land at once against a fixed deadline: four model boards have to stitch into one shell, and roughly 7,300 words of proposal must compress far enough for results to fit inside 8,500.

#### NEXT WEEK

Stitch all FORGE components into one site; finish and submit the final paper; turn in, present, freeze the repos.